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标题	Diastereoselective Synthesis of N-Methylspiroindolines by Intramolecular Mizoroki – Heck Annulations
作者	Lindman J., Gopalan G., Palo-Nieto C., Brandt P., Gising J., Larhed M.
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摘要（原文节选）

Spiroindolines represent a privileged structure in medicinal chemistry, although stereocontrol around the spirocarbon can be a synthetic challenge. Here we present a palladium(0)-catalyzed intramolecular Mizoroki – Heck annulation reaction from (+)-Vince lactam-derived cyclopentenyl-tethered 2-bromo-N-methylanilines for the formation of N-methylspiroindolines. A series of 14 N-methylspiroindolines were synthesized in 59 – 81% yield with diastereoselectivity >98%, which was rationalized by density functional theory c ⋯

数据表 · Table 1（15 行，含数值行 15 行）

Optimization of Reaction Conditions for Intramolecular Mizoroki – Heck Spirocyclization of 6/8aa

entry	starting material	Pd (mol %)	ligand (mol %)	temp (° C)	time (h)	NMR ratio (6:7 or 8a:9a)b	isolated yield of 7/9ac
1	6	Pd(OAc)2 (5)	dppf (10)	80	48	11:89	
2	6	Pd(OAc)2 (5)	dppf (10)	100	16	25:75	
3	6	Pd(OAc)2 (5)	dppf (10)	120	5	19:81	
4	6	Pd(OAc)2 (5)	dppf (10)	120	16	22:78	
5	6	Pd(OAc)2 (5)	dppf (5)	80	5	83:17	
6	6	Pd(OAc)2 (5)	dppf (5)	80	16	82:18	
7	6	Pd(dppf)Cl2 (5)		80	16	90:10	
8	6	Pd(OAc)2 (5)	PPh3 (10)	80	16	55:45	
9	6	Pd(OAc)2 (5)	XPhos (10)	80	16	96:4	
10	6	Pd(t-Bu3P)2 (5)		80	5	11:89	
11	6	Pd(t-Bu3P)2 (10)		80	16	1:99	64%
12	6		t-Bu3P (10)	80	16	100:0	
13d	6	Pd(t-Bu3P)2 (10)		80	16	97:3	
14	8a	Pd(t-Bu3P)2 (10)		80	16	1:99	83%
15	8a	Pd(OAc)2 (5)	dppf (10)	80	16	47:53	

脚注（逐字）：aReaction conditions: cyclization precursor 6 or 8a (0.1 mmol, 1 equiv), Et3N (2 equiv), DMF. Amounts of palladium precatalyst and ligands listed in the table.bRatio between 6:7 or 8a:9a determined by 1H NMR of crude reaction mixture.cIsolated yields after column chromatography.dExperiment run witho

正文对应句（逐字）：With substrate 6 in hand, screening of reaction conditions for the Mizoroki – Heck cyclization was commenced. The results are presented in Table 1. A Pd(OAc)2/dppf catalytic system with Et3N as the base and DMF as the solvent was chosen as the starting point, as this has previously been successfully u

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